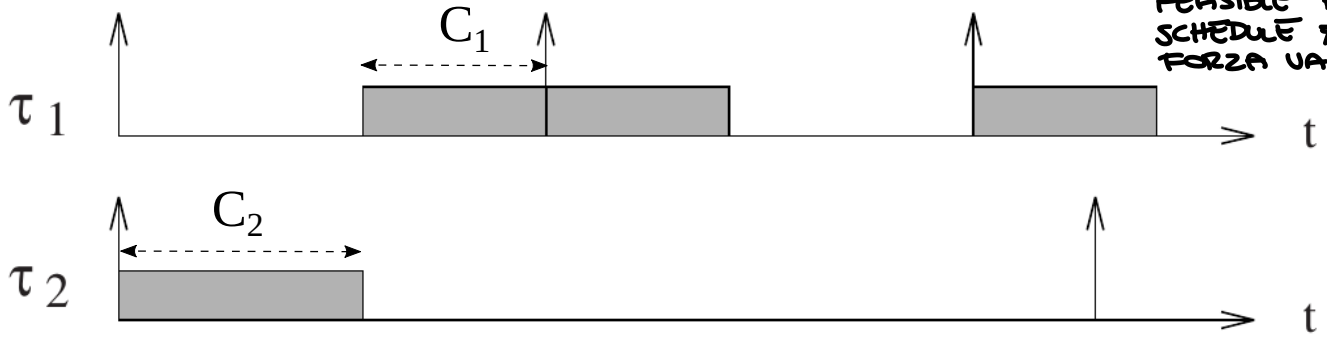
RM: proof of optimality for a task set made of 2 tasks (1/3)

- Consider a task set Γ made of 2 periodic tasks τ_1 and τ_2 with $T_1 < T_2$ (the proof can be easily extended to a task set made of n tasks)
- If priorities are not assigned according to RM $\Rightarrow$ priority τ_2 > priority τ_1 $\Rightarrow$
  $\Rightarrow$ The schedule is feasible for Γ at the critical instants if $\underbrace{C_1 + C_2 \leq T_1}$

SUPPOSTO CHE T SIA FEASIBLE PER QUESTO SCHEDULE ≠ RM, DEVE PER FORZA VALERE QUESTA



Remark

It is sufficient to prove that:

If $C_1 + C_2 \leq T_1 \Rightarrow$ The RM schedule is feasible for Γ at the critical instants

Image adapted from "Hard real-time computing systems" by Prof. G. Buttazzo

RM: proof of optimality (2/3)

- Number of periods of τ_1 entirely contained in T_2: $F := \lfloor T_2/T_1 \rfloor$
- If priorities are assigned according to RM $\Rightarrow$ priority τ_1 > priority τ_2
- Case 1 (left): $C_1 < T_2 - F\,T_1$ (i.e., all the jobs of τ_1 released within $[0, T_2)$ are completed before the second job of τ_2 is released)

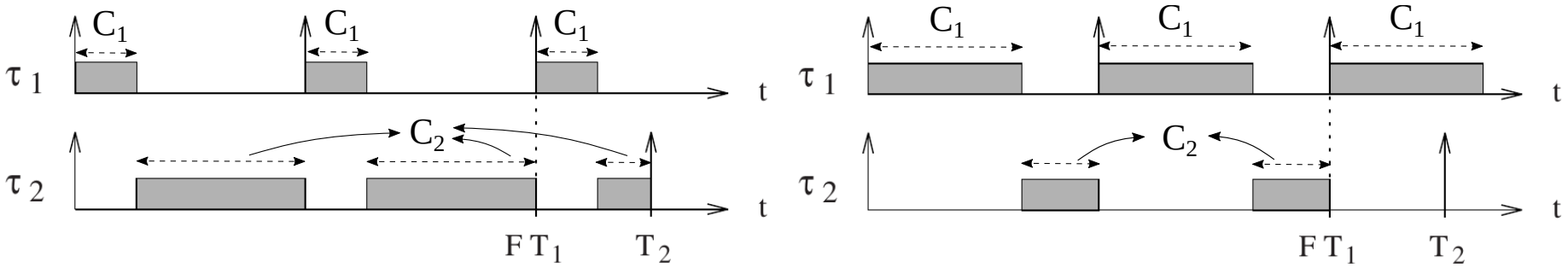
  $\Rightarrow FT_1 + C_1 < T_2 \Rightarrow$ GLI F JOB + UN ALTRO DEVONO FINIRE PRIMA CHE T_2 SIA ATTIVATO AL TEMPO T_2.

  - The task set is schedulable FOR RM if $(F+1)C_1 + C_2 \le T_2$

    $\Rightarrow$ IN $[0, T_2]$ C'È UN JOB DEL TASK τ_2 E F+1 JOB DEL TASK τ_1 (UN TASK PER OGNI PERIODO T_1 PIÙ QUELLO CHE INIZIA AL TEMPO FT_1).
  - We prove that $C_1 + C_2 \le T_1 \Rightarrow (F+1)C_1 + C_2 \le T_2$

    $\Rightarrow$ DIMOSTRO IL REMARK PAGINA SOPRA.
- Case 2 (right): $C_1 \ge T_2 - F\,T_1$ (i.e., some job of τ_1 released within $[0, T_2)$ is not completed before the second job of τ_2 is released)
  - The task set is schedulable if $F\,C_1 + C_2 \le F\,T_1$

    $\Rightarrow$ NON HA SENSO CONSIDERARE T_2 PERCHÉ NON SAPPIAMO AL TEMPO T_2 QUANTO IL JOB DI τ_1 HA ESEGUITO. ALLO RA SI CONSIDERA F VOLTE IL TEMPO T_1.
  - We prove that $C_1 + C_2 \le T_1 \Rightarrow F\,C_1 + C_2 \le F\,T_1$



Images adapted from "Hard real-time computing systems" by Prof. G. Buttazzo

RM: proof of optimality (3/3)

- Case 1: We prove that $C_1 + C_2 \leq T_1 \Rightarrow (F+1)C_1 + C_2 \leq T_2$
  - $C_1 + C_2 \leq T_1 \Rightarrow F\,C_1 + FC_2 \leq F\,T_1$ given that $F \geq 1 \Rightarrow$ VERO PERCHÉ $T_2 \geq T_1$
    $\Rightarrow F\,C_1 + C_2 \leq F\,C_1 + F\,C_2 \leq F\,T_1$ given that $F \geq 1 \Rightarrow$
    $\Rightarrow (F+1)C_1 + C_2 \leq F\,T_1 + C_1 \Rightarrow$
    $\Rightarrow (F+1)C_1 + C_2 \leq F\,T_1 + C_1 < T_2$ given that $C_1 < T_2 - F\,T_1 \Rightarrow$
    $\Rightarrow (F+1)C_1 + C_2 < T_2$

- Case 2: We prove that $C_1 + C_2 \leq T_1 \Rightarrow FC_1 + C_2 \leq FT_1$
  - $C_1 + C_2 \leq T_1 \Rightarrow F\,C_1 + FC_2 \leq F\,T_1$ given that $F \geq 1 \Rightarrow$
    $\Rightarrow F\,C_1 + C_2 \leq F\,C_1 + F\,C_2 \leq F\,T_1$ given that $F \geq 1 \Rightarrow$
    $\Rightarrow F\,C_1 + C_2 \leq F\,T_1$